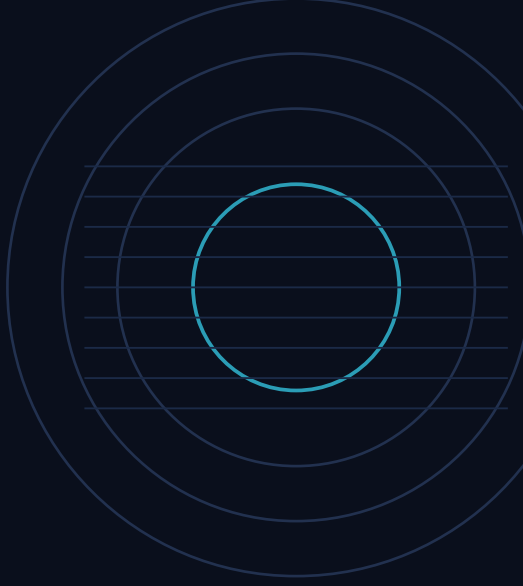


THIN FILM ENGINEER

The Semiconductor Process Interview Pack

Fifty questions actually asked in process-engineering interviews — deposition, plasma etch, vacuum, metrology, statistics and troubleshooting — each with a model answer that leads with mechanism, not memory. Plus five one-page reference sheets to keep open next to the tool.

50 Q&A · PVD · ALD · CVD · ETCH · SPC

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HOW TO USE THIS PACK

Answer with mechanism, not memory.

Every model answer below follows the same skeleton: **mechanism first** (what is physically happening), **then the knob** (what you would change and in which direction), **then the caveat** (what you would check before believing yourself). Interviewers reward that order. Memorized facts sound like facts; mechanisms sound like experience.

Don't recite these answers. Read one, close the page, and say your own version out loud — shorter is better. The answers are deliberately fuller than what you'd say live, so you can see the reasoning; a live answer is the first two or three sentences plus whatever the follow-up digs into.

The five-step loop the troubleshooting questions use: **01 OBSERVE** what exactly changed · **02 SPLIT** tool / process / metrology · **03 HYPOTHESISE** a mechanism, not a guess · **04 DISCRIMINATE** one test, two outcomes · **05 VERIFY & LOCK** prove it, document it. Step 04 is where most people fail: a good test rules something out whichever way it comes back.

The reference sheets at the back are one page per topic on purpose — print them, keep them at the tool, annotate them with your own fleet's numbers.

SECTION 01

PVD & SPUTTERING

How films are knocked off a target and land on your wafer

Q01 Walk me through what happens in a sputtering chamber from process start to film on the wafer.

Argon flows in and the chamber sits at a few millitorr. Applied power ignites a plasma; the magnetron's field traps electrons near the target, densifying it. Ar+ ions accelerate across the target sheath and arrive with hundreds of eV, knocking neutral atoms off the target by momentum transfer. Those atoms fly through the gas — scattering more at higher pressure — land on the wafer, skate along the surface, nucleate islands, and the islands coalesce into a continuous film. Thickness comes from power and time; uniformity from geometry and pressure; stress and density from the energy of arriving species.

Q02 Your film is 8% thicker at the wafer edge. What are the likely causes, and what do you change first?

First separate tool from process: is the profile new (something changed) or chronic (geometry)? Edge-thick usually points to target erosion-track shape, target-to-substrate spacing, or a pressure change that altered the arrival-angle distribution. The cheapest knob is usually pressure — more scattering randomizes arrival directions and flattens the profile — followed by spacing. Check target life first: a deeply eroded racetrack redistributes flux and no recipe knob fully compensates. If the tool has them, tune per-zone deposition time or use a uniformity shaper.

Q03 What does raising chamber pressure do to a sputtered film, mechanistically?

Higher pressure shortens the mean free path, so sputtered atoms suffer more collisions in transit. They arrive with lower energy and from more randomized angles. Consequences: better sidewall/step coverage but worse bottom fill, lower film density, a shift from compressive toward tensile stress (less atomic peening), and usually a lower deposition rate. Almost every pressure question is really a mean-free-path question — answer it that way.

Q04 DC, pulsed-DC and RF sputtering — when do you use each?

DC is the workhorse for conductive targets: simple, fast, cheap. Insulating targets charge up and arc under DC, so RF (typically 13.56 MHz) alternates the polarity faster than charge can build, at the cost of rate and hardware complexity. Pulsed-DC is the middle ground for reactive processes on conducting targets (e.g. AlN, TiO_2 from metal targets): brief reverse pulses discharge the poisoned, insulating regions before they arc.

Q05 What is target poisoning in reactive sputtering, and how do you run a stable process despite it?

In reactive sputtering the reactive gas (N_2, O_2) doesn't just react at the wafer — it reacts with the target face, forming a compound layer whose sputter yield is far lower. Rate collapses, and the metal-to-poisoned transition is hysteretic: the flow at which you fall into poisoned mode is higher than the flow needed to climb back out. Stable operating points sit on the transition edge, so production tools hold them with feedback — controlling on cathode voltage, optical emission or partial pressure rather than on a fixed gas flow.

Q06 Where does stress in a PVD film come from, and how do you tune it?

Intrinsic stress comes from how the film is assembled: energetic bombardment hammers atoms into the lattice (atomic peening, compressive), while low-mobility growth leaves under-dense grain boundaries that pull the film tight (tensile). Extrinsic stress comes from thermal-expansion mismatch on cool-down. The practical knobs are pressure (more scattering → less peening → tensile shift), bias and power (more ion energy → compressive shift), and temperature. State the sign convention out loud in an interview — film that wants to shrink is tensile, wafer bows toward the film.

Q07 Step coverage in a deep via is poor. What can you do within PVD — and when do you give up and use ALD?

PVD is line-of-sight, so the fixes are geometric: collimation or long-throw at low pressure to favor normal-incidence flux for the bottom, ionized PVD (iPVD) with wafer bias to draw metal ions vertically, or resputter to redistribute bottom material onto sidewalls. Each trades rate and adds overhang risk at the via mouth. Beyond roughly 5:1 aspect ratio, or when you need continuous nm-thin liners on all surfaces, stop fighting geometry and use ALD — that's what it's for.

Q08 A film that adhered fine last week now peels. Where do you look first?

Adhesion failures are interface stories, so interrogate what the first monolayer saw: surface prep (was a preclean or degas skipped or weakened?), vacuum quality (water partial pressure at deposition start), time-between-steps (queue time lets surfaces re-adsorb contaminants), and stress (did thickness or recipe drift push stored elastic energy past the interface strength?). The Monday-adheres-Friday-peels pattern is classically a time-dependent variable — queue time, chamber idle state, or humidity — not a recipe change. Check the logs before the recipe.

SECTION 02

ALD & CVD

Films grown by chemistry — one layer at a time, or continuously

Q09 Explain one ALD cycle. Why is growth self-limiting?

One cycle is four steps: dose precursor A, purge; dose co-reactant B, purge. The precursor chemisorbs only on reactive surface sites; once they're occupied, further precursor doesn't stick — the surface reaction shuts itself off. The co-reactant then regenerates the sites. Because each half-reaction saturates, you get a fixed growth-per-cycle (~0.5–1.5 Å) regardless of small dose variations, which is why thickness control is digital (count cycles) and conformality is near-perfect.

Q10 What is the ALD temperature window, and what happens on each side of it?

It's the temperature range where growth-per-cycle is flat because both half-reactions saturate cleanly. Too cold: precursor physisorbs or reactions are incomplete — GPC rises spuriously (condensation) and films go impure. Too hot: the precursor decomposes thermally, adding a parasitic CVD component (GPC rises, conformality and control degrade) — or surface species desorb and GPC falls. A flat GPC-vs-T plateau is one of the standard proofs you're doing real ALD.

Q11 How do you prove your process is true ALD and not pulsed CVD?

Run the saturation curves: GPC versus precursor dose time and versus co-reactant dose time must plateau — more dose, same growth. GPC versus purge time must also plateau; if shortening purges raises GPC, precursors are meeting in the gas phase (CVD component). Add the temperature-window check and linearity of thickness versus cycle count through zero. Conformality in a high-AR test structure is the final witness.

Q12 CVD has surface-reaction-limited and mass-transport-limited regimes. Why does the distinction matter?

Rate is set by the slower of two series steps: reactant delivery through the boundary layer, or reaction at the surface. At lower temperature the reaction is slow and rate-limiting — rate follows Arrhenius, so you need excellent temperature uniformity, but gas-flow non-uniformity is forgiven (batch furnaces live here). At higher temperature delivery limits — rate depends on flow and boundary-layer geometry, so reactor design and depletion along the flow path dominate uniformity. Every CVD uniformity problem starts with 'which regime am I in?'

Q13 PECVD versus thermal CVD — what does the plasma buy you and what does it cost?

The plasma cracks precursors with electron impact instead of heat, so you deposit at 200–400 °C instead of 600 °C+ — essential once metals or finished devices are on the wafer. You also gain a stress knob (ion bombardment, including dual-frequency control) and higher rates. The costs: more hydrogen and impurities in the film, possible plasma damage to sensitive layers, poorer conformality than thermal processes, and film properties that now depend on RF parameters as well as chemistry.

Q14 What goes wrong when ALD purge times are too short?

Residual precursor A meets incoming co-reactant B in the gas phase or on chamber walls — a parasitic CVD reaction. Symptoms: growth-per-cycle rises above the literature value and drifts, within-wafer uniformity and conformality degrade, particles appear (gas-phase nucleation), and the film picks up impurities. If throughput pressure tempts you to shave purges, verify with the purge-saturation curve, not hope.

Q15 Why is ALD conformal in ways CVD and PVD can't be — and what still fails in extreme aspect ratios?

Because saturation makes flux irrelevant: the bottom of a trench sees less precursor per second than the top, but as long as it eventually saturates during the dose, it grows the same monolayer fraction. PVD needs line-of-sight and CVD rate follows local flux, so both thin out downhole. In extreme structures ALD still fails when doses can't reach saturation at the bottom (insufficient exposure — Knudsen transport limited), when purges can't clear the depth (CVD component downhole), or when the precursor decomposes or recombination consumes the co-reactant (common for plasma ALD radicals) before reaching the bottom.

Q16 After a chamber wet-clean, your first CVD lots show particles. What's your reasoning?

Fresh chamber surfaces behave differently from seasoned ones: adhesion of deposits to bare walls is poor, wall temperature and chemistry shifted, and the first films to build on walls can flake. Standard practice is seasoning — deposit a sacrificial coating on the clean chamber before production. If particles persist: check for redeposited flakes from the clean itself, verify the season recipe ran complete, then look at gas-line moisture (opened lines) driving gas-phase nucleation. Particle maps help — flaking gives clustered/streaked patterns, nucleation gives uniform scatter.

SECTION 03

PLASMA ETCH

Taking material away — straight down, and only where you mean to

Q17 In reactive ion etching, what does the actual etching — ions or radicals?

Both, and the synergy is the point. Radicals chemisorb and form volatile products but etch slowly and isotropically on their own; ions sputter slowly on their own. Together — chemistry weakening the surface plus directional ion energy driving product formation and desorption — the vertical etch rate can exceed the sum of both (the classic Coburn–Winters result). Anisotropy exists because ions arrive vertically through the sheath while radicals arrive from all angles; sidewalls see chemistry but no bombardment, and passivation keeps them from etching.

Q18 Name the main etch knobs and what each really does.

Source power sets plasma density → radical and ion flux → rate. Bias power sets ion energy → anisotropy, mask erosion, damage. Pressure sets mean free path and residence time → ion angular spread and the radical-to-ion balance. Gas chemistry sets the etch/passivation balance (e.g. F:C ratio in fluorocarbons; O_2 leans it toward etch, CH-additions toward polymer). Individual flows tune selectivity and profile; wafer temperature controls passivation sticking and product volatility; time is depth. Answer any profile question by naming which of these moved.

Q19 The trench profile is tapered (sloped inward). What's happening and which direction do you correct?

Taper means the sidewall is over-protected or the bottom under-attacked: too much passivation, too little ion energy, or too much ion-angle scatter. Corrections: lean the chemistry away from polymer (more O_2, less CH-rich gas), raise bias power, or lower pressure to tighten the ion angular distribution. Also rule out the mundane: a sloped or faceted mask edge prints as a sloped feature — check the resist profile before touching plasma knobs.

Q20 Now the profile is undercut and bowed. Same question.

Undercut is unopposed lateral chemistry: passivation is failing or radicals dominate ions. Add or strengthen the passivant, drop pressure or wafer temperature (passivation sticks better cold), or raise bias so the bottom clears before sidewalls soak. Bowing specifically implicates ions deflected into the sidewall — scattered at high pressure, or refracted by charging in the feature — and mask faceting that opens a wider acceptance angle mid-depth. State the mechanism, then the knob; that order is what interviewers listen for.

Q21 What is selectivity and how do you raise etch-to-mask selectivity without wrecking the profile?

Selectivity is the rate ratio between the film you're etching and the material you must preserve (mask above, stop-layer below). Chemical routes: shift chemistry so byproducts are volatile for the film but not the mask — in fluorocarbon etches, richer C:F chemistry deposits protective polymer on resist and oxide-selective steps exploit the polymer forming on Si but not SiO_2 under bombardment. Physical route: drop bias, since sputtering is unselective — but you pay in profile. Real recipes ride the polymerizing edge: maximum selectivity just before etch-stop.

Q22 Why do narrow features etch slower than wide ones (RIE lag / ARDE)?

Transport into the feature degrades as aspect ratio grows: neutrals arrive by molecular flow with a conductance that shrinks with AR, off-vertical ions are shadowed by the mouth, and positive charging at the insulating bottom decelerates and deflects incoming ions. All three starve the bottom of a narrow trench relative to a wide one. Mitigations: higher bias, lower pressure, pulsed plasmas to let charge dissipate — or switch the critical step to ALE, which removes ARDE by making removal-per-cycle flux-independent.

Q23 How do you know when the etch is done? Compare endpoint methods.

Optical emission spectroscopy watches plasma lines from etch products or reactants — when the film clears, product lines fall and reactant lines rise; it's the workhorse but needs enough open area to see a signal. Laser interferometry watches reflectance oscillations on the actual film, giving real-time rate and remaining thickness on transparent films. RF impedance shifts also flag material transitions. Low-open-area etches are the hard case: signal averaging, multivariate models, or timed etch on a rate qualified by a monitor wafer. Always follow endpoint with a controlled overetch — and know what your overetch percentage costs in stop-layer loss.

Q24 Why do ICP tools separate source power from bias power?

One knob cannot serve two masters: in a simple capacitive tool, RF power sets both plasma density (flux) and sheath voltage (ion energy) together. An ICP's coil couples power inductively to make a dense plasma with a low natural sheath voltage, while an independent RF bias on the chuck sets ion energy. You can run high flux with gentle ions (fast but low-damage), or modest flux with energetic ions, independently — that decoupling is most of why high-density-plasma etchers took over anisotropic etch.

Q25 How does atomic layer etching work, and when is it worth its slowness?

It's the etch mirror of ALD: a self-limiting modification step (e.g. Cl_2 chlorinates one surface layer of Si), purge, then a gentle removal step whose ion energy sits in a window — enough to remove the modified layer, not enough to sputter pristine material — then purge, repeat. Removal per cycle is fixed, so you get atomic-scale depth control, smooth surfaces, near-zero ARDE and superb selectivity. It's slow, so you spend it only where angstroms decide yield: gate-recess landings, thin channels in GAA structures, 2D materials.

SECTION 04

PLASMA & VACUUM FUNDAMENTALS

The physics the fab assumes you already know

Q26 What is a plasma sheath, and why do ions arrive at the wafer near-vertically?

Electrons are far faster than ions, so any surface touching plasma initially collects more electrons and charges negative until it repels most of them — leaving a thin, positive, electron-depleted region: the sheath. The resulting field points from plasma to surface, so ions falling through it accelerate along the surface normal. Their trajectories, randomized in the bulk, get collimated by the sheath — arriving within a few degrees of vertical at low pressure. That collimation is the entire basis of anisotropic etching.

Q27 Base pressure won't recover after a chamber open. Give me your systematic search.

First separate real leak from virtual load with a rate-of-rise: isolate the chamber and log pressure versus time. A real leak climbs linearly and doesn't care about pump-time history; outgassing (water) starts fast, decays with time, and improves with heat. If real: check what was touched — last-opened seals first, look for pinched or dusty O-rings, then helium leak-check joints methodically from most-recently-disturbed outward. If outgassing: bake, purge with warm dry gas, and review what was put inside (fingerprints, wrong cleaning solvent, trapped volumes). Most post-PM 'leaks' are water.

Q28 Why does mean free path matter so much, and how does it scale?

Mean free path is the average distance between collisions; it's inversely proportional to pressure — roughly 5 cm at 1 mTorr for air-like gases, 0.05 mm at 1 Torr. It decides whether transport is ballistic (line-of-sight PVD, collisionless sheaths, tight ion angles) or diffusive (scattered arrival, conformal-ish coverage, spread ion angles). When λ is much larger than the chamber you reason with geometry; when smaller, with flow and diffusion. Enormous numbers of process questions reduce to 'which side of that comparison are you on?'

Q29 Walk me through the pump stack on a process tool and why each stage exists.

Roughing pumps (rotary vane, scroll, multi-stage dry roots) work from atmosphere down to ~10■²–10■³ Torr; they move bulk gas but can't establish molecular flow vacua. Turbomolecular pumps take over below that, imparting momentum with 30–60 krpm blades down to 10■■–10■■ Torr — but they can't exhaust to atmosphere, hence the backing pump behind them. Cryopumps capture gas on cold arrays — enormous water pumping speed, ideal for PVD load-locks — but store what they pump and need regeneration. Know which pump handles corrosive etch flows (hardened dry pumps, heated lines to stop condensation) and why turbos on etch tools get purged bearings.

Q30 How do you tell outgassing (or a virtual leak) from a real leak?

Same tool, three signatures. Rate-of-rise shape: linear and history-independent means real leak; decaying with pump time means desorption. Response to heat: baking accelerates outgassing sharply but barely changes a real leak. Composition, if you have an RGA: air leak shows N_2:O_2 near 4:1 plus argon; outgassing is dominated by water; a virtual leak (trapped volume — unvented screw, double O-ring) looks like outgassing but doesn't respond to bake, because the gas reservoir refills the chamber through a tiny conductance regardless of wall temperature. Vented hardware exists precisely for this.

Q31 What sets ion energy at the wafer, and what turns a sharp energy into a spread?

The time-averaged voltage across the wafer sheath — self-bias plus applied RF bias — sets the center energy. Two things smear it: RF modulation (ions crossing a sheath whose voltage swings during their transit arrive with a bimodal spread; heavier ions average more, lighter ions split wider) and collisions in the sheath at higher pressure (charge exchange creates slow ions and off-axis velocities). Low pressure + high frequency bias → tight, near-vertical, well-defined energy; that's the regime precision etch lives in.

Q32 How does a mass flow controller work, and how does a drifting one show up?

A thermal MFC splits flow past a heated sensor tube; gas carrying heat downstream shifts a temperature difference proportional to mass flow, and a valve servos to setpoint. It's calibrated per-gas (gas correction factors) and can read wrong if clogged, if the bypass ratio shifts, or if inlet pressure moves outside spec. A drifting MFC is nasty because the readback follows the sensor — the tool says the flow is fine while the true flow walks. Suspect it when chemistry-sensitive symptoms (rate, selectivity, uniformity) drift with no recipe change; confirm with a rate-of-rise flow verification into a known volume, which most modern tools automate.

SECTION 05

METROLOGY

Measuring what you made — and doubting it correctly

Q33 How does ellipsometry measure a film you can't even see?

It reflects polarized light off the stack and measures how the polarization changed — the amplitude ratio (Ψ) and phase shift (Δ) between p- and s-components — versus wavelength. Those depend on every layer's thickness and optical constants, so you fit an optical model to the measured spectra; the fitted parameters are your answer. It's fast, non-contact and sub-nm repeatable. The catch: you measure a model, not a film — remember that phrase.

Q34 When does ellipsometry lie to you?

Whenever the model is wrong: unknown or wrong n & k (thickness and index correlate strongly for thin layers), missing interlayers or native oxide, surface roughness modeled as a sharp interface, thickness outside the sensitivity range, or backside reflections on transparent substrates. Watch the fit quality (MSE) and parameter correlations, and sanity-check against an independent technique — XRR or TEM — when the film is new. If the recipe's model was built for the old film and the process changed the film's density or composition, the 'thickness drift' you're chasing may be pure fiction.

Q35 What does XRR give you that ellipsometry doesn't?

X-ray reflectivity fringes depend on electron density, thickness and interface roughness — with essentially no optical-constant assumptions. So it gives density directly (a film-quality readout ellipsometry can't provide), works on metals and opaque-in-visible films, and separates roughness from thickness. It's slower and needs flat blankets, so the standard play is XRR to anchor truth and calibrate the ellipsometer model, ellipsometry for the high-throughput fleet monitor.

Q36 How do you measure film stress in production, and what's the underlying relation?

Measure wafer curvature before and after deposition — laser scan across a diameter — and put the curvature change into Stoney's equation with substrate modulus and thickness and film thickness. Sign convention: film that wants to contract bows the wafer toward itself (tensile, concave film-side). Assumptions worth naming: thin uniform film on a much thicker substrate, and it reports the average biaxial stress, not local or gradient stress. Track stress×thickness through a stack to predict bow at lithography.

Q37 You're handed a bad SEM cross-section. What do you actually look at?

Read it like a flight recorder, top down: mask condition first (faceted? eroded? sloped resist prints a sloped feature), then CD at top/middle/bottom to classify taper–bow–undercut, then the foot or notch at the stop layer, sidewall texture (striations = mask edge roughness or passivation cycling), and any residue or micro-trenching at the corners. Each signature maps to a mechanism — facet → mask selectivity; bow → scattered/deflected ions; notch → charging at the insulator. Also check the sample prep: a tilted or smeared cleave manufactures artifacts that look like process problems.

Q38 What does AFM give you, and what artifacts must you respect?

A nm-sharp tip on a cantilever traces the surface, giving true 3-D topography and sub-Å-class roughness numbers (Ra, Rq, PSD) that SEM can't quantify. The core artifact is tip convolution: every image is the surface dilated by the tip shape — narrow trenches read shallow, sharp features read tip-shaped, and a worn or contaminated tip 'changes your process' overnight. Trust roughness trends only with a tip-check cadence, and treat any sudden roughness step that coincides with a tip change as metrology until proven otherwise.

Q39 A monitor's thickness reading drifted. Design the test that tells you process vs metrology in one move.

Freeze a golden wafer — one measured many times, known stable — and remeasure it now. If the golden wafer moved, the metrology moved: recalibrate, check lamp/alignment/tip, look for a model mismatch. If the golden wafer is stable but product drifted, the process really moved — go hunt the tool. That's the general pattern interviewers want: one experiment whose two outcomes cleanly split the hypothesis space. Gauge R&R; is the institutional version — never chase a process signal smaller than your measurement noise.

Q40 What is Gauge R&R; and when do you insist on one?

A designed study splitting measurement variation into repeatability (same operator, same setup, repeated) and reproducibility (across operators/loads/days), reported against the tolerance or the process spread — %GRR under ~10% is healthy, over 30% means the gauge can't see what you're controlling. Insist on it before qualifying a new measurement, before tightening specs, and before any drift investigation with stakes: chasing process ghosts with a noisy gauge burns weeks. It's also the honest answer to 'how do you know your 42.3 nm is 42.3?'

SECTION 06

PROCESS STATISTICS — SPC, CAPABILITY & DOE

The language decisions are made in

Q41 Explain Cp and Cpk to a manager, then tell me what Cpk = 1.02 means and what you'll do.

Cp asks: is the process spread narrow enough to fit between the spec limits? Cpk asks the same but also charges you for being off-center — it's the distance from the mean to the nearer spec limit in units of three sigma. Cpk 1.02 means the process edge is essentially touching a spec limit: roughly a 0.1% fallout tail today and no margin for any drift. Plan: first verify the gauge (GRR), then recenter if the mean is off — centering is cheap — and if the spread itself is the problem, run a DOE to find the variance drivers. Interim: tightened monitoring or screening until Cpk recovers to the house target (typically ≥1.33).

Q42 Control limits versus spec limits — why must they never be confused?

Control limits are the voice of the process: computed from its own variation (±3σ of the plotted statistic), they say whether it's behaving as it always has. Spec limits are the voice of the customer: what the product needs. A process can be in control and out of spec (stable but incapable), or in spec but out of control (still passing while something has changed — your early warning). Drawing spec limits on a control chart and 'controlling' to them guarantees both false alarms and missed drifts — it's the most common SPC abuse in fabs.

Q43 What signals on a control chart say 'act' besides a point outside the limits?

Non-random patterns inside the limits: runs of 7–8 points one side of center (mean shift), steady trends up or down (drift — target life, buildup), 2-of-3 near a limit (Western Electric zone rules), and cycling (alternating chambers or shift patterns — stratification). Each pattern hints at its cause class: sudden jump → discrete event (PM, lot change); trend → consumable; cycles → a rotating variable. The chart isn't a pass/fail gate; it's a diagnostic waveform.

Q44 Your process is in control but not capable. What are your options, honestly ranked?

In control means it's doing its stable best; not capable means its best isn't good enough. Options: (1) recenter if off-target — nearly free; (2) reduce variance — find dominant contributors (nested variance analysis: within-wafer vs wafer-to-wafer vs lot-to-lot points at different mechanisms) and DOE the drivers; (3) re-examine the spec — some limits are inherited, not derived, and a Cpk crisis against a fictional spec is a paperwork problem; (4) live with screening — 100% measurement and disposition — which is a treadmill, not a fix. Saying '2 with 1 first, while auditing 3' is the senior answer.

Q45 Design an experiment to improve thickness uniformity. Why factorial rather than one-variable-at-a-time?

Frame it: response = within-wafer 1σ; suspected factors from mechanism — say pressure, spacing, power, temperature. Screen with a 2-level fractional factorial (e.g. 2■■¹ = 8 runs plus 3 center points), randomized run order, replicated centers for pure error and curvature check. OFAT fails for exactly the reason uniformity problems are hard: interactions — the best pressure at one spacing is the wrong pressure at another, and OFAT walks to a false optimum along one axis at a time. The factorial estimates all main effects and interactions with the same runs, at a known confidence. Follow the screen with a response-surface design around the winners.

Q46 What is confounding in a fractional factorial, and why add center points?

Fractionating means some effects share the same contrast — a main effect may be aliased with an interaction (in a Resolution III design, with two-factor interactions; Resolution IV pushes that to three-factor). You buy fewer runs with ambiguity you must resolve later — by fold-over or targeted follow-ups — if an aliased string comes up significant. Center points do two jobs: replicated pure-error estimate for honest significance testing, and curvature detection — if the center response falls off the plane predicted by the corners, the world isn't linear and you need a quadratic (response-surface) design before optimizing.

SECTION 07

TROUBLESHOOTING & THE HUMAN SIDE

The questions that actually decide senior hires

Q47 Etch rate dropped 20% overnight with no recipe change. Run your loop out loud.

Observe: what exactly changed and when — one chamber or all, step change or drift, which layer/recipes affected, what events bracket the change (PM, gas bottle swap, part replacement, new mask reticle)? Split: tool, process or metrology — is the rate really down, or is the measurement off (golden-wafer check)? Hypothesize from mechanism: rate ∝ radical flux and ion energy, so suspects are RF delivery (reflected power, matching), gas delivery (MFC drift, bottle purity), pressure reading (gauge), wall condition (first lots after clean), leaks (O_2/H_2O scavenging F). Discriminate: pick the test that splits the space fastest — the tool's own logs first (reflected power, DC bias, valve positions, endpoint traces cost zero wafers), then one monitor wafer if needed. Verify & lock: reproduce the fix, run SPC monitor, write it down. Delivering that structure calmly is the interview.

Q48 The film adheres on Monday and peels on Friday. Find the variable.

The recipe didn't change with the day of the week, so something time-correlated did. List what cycles weekly: queue time between clean/deposition steps (Friday backlog?), chamber idle time and wall state, DI or solvent bath age, ambient humidity, gas bottle depletion, even which crew runs the step. Pull the peeling lots' histories and diff them against Monday lots — the discriminating variable is usually sitting in the timestamps, not the recipe. Then confirm causally: force the suspect condition on purpose (age a queue deliberately) and watch peel follow. That last step — recreating the failure — is what separates a fix from a coincidence.

Q49 Tell me about a time your root cause was wrong. (How to answer it.)

Pick a real story where you committed to a cause, acted, and the problem returned — then tell it in the loop's language: what evidence fooled you (correlation posing as mechanism), what discriminating test you should have run first, how you recovered, and what changed in your method afterward — e.g. 'I now write the falsifying test before I write the fix.' Interviewers ask this to find engineers who update on evidence rather than defend their first guess. Never pick a story where you were secretly right all along; that answers a different, worse question.

Q50 A tool comes back from major maintenance. How do you prove it's healthy before releasing it to production?

Staged release, cheapest checks first: hardware state (leak-up rate, base pressure, RF match ranges, temperature and flow verifications against the PM checklist), then plasma/process fingerprints without product (ignition, bias voltage, OES signature versus golden traces), then qualification wafers — rate, uniformity, particles, and the film metric that tool owns — against control limits, not just specs. Then conditional release: first N production lots at tightened sampling with SPC watch. The principle to say out loud: release is a statistical statement — 'this tool is indistinguishable from its known-good self' — not a feeling that the PM went fine.

Q51 Explain a complex process problem to a non-technical manager without dumbing it down.

Lead with what they own: impact and decision — 'we're at risk of holding N lots; I need two days and four monitor wafers to be sure of the fix.' Then one sentence of mechanism, translated: not 'the sheath collapsed,' but 'the layer isn't being compacted as it grows, so it's less dense and etches too fast.' Then options with costs and your recommendation. Numbers in their currency: Cpk 1.02 → 1.41 becomes 'we can stop inspecting every lot, which returns a tool-day per week.' Close with the date they'll hear from you next. Precision about impact is respect; jargon is not rigor.

REFERENCE SHEET

PVD SHEET — SPUTTERING AT A GLANCE

THE ONE MENTAL MODEL

– Everything is momentum transfer + transport: ions knock atoms off the target; pressure decides how they fly; substrate condition decides how they land.

– Mean free path λ ∝ 1/P. λ ■ throw distance → ballistic, directional, dense, compressive. λ ■ throw → diffusive, conformal-ish, porous, tensile.

KNOB → EFFECT (FIRST-ORDER)

– Power ↑ → rate ↑ (≈linear), slight energy ↑ · Pressure ↑ → scattering ↑, density ↓, stress → tensile, step coverage sidewalls ↑ / bottom ↓.

– Spacing ↑ → uniformity ↑, rate ↓ · Bias ↑ → densification, compressive shift, resputter (mouth clipping in vias).

– Temperature ↑ → adatom mobility ↑ → larger grains, lower resistivity, stress relaxation.

STRUCTURE-ZONE SHORTHAND

– Low T/Tm, low energy → Zone 1 porous columns · add energy (bias/ions) → Zone T dense fibrous · higher T/Tm → Zones 2–3 columnar→equiaxed.

– Use it to answer any 'why is my film rough/porous/stressed' question in one breath.

TROUBLESHOOTING QUICK TABLE

– Edge-thick → target erosion / spacing / pressure · Peeling → interface prep, water partial pressure, stress·thickness · Arcing → poisoned target areas, damaged shields.

– Rate down at same power → target near end-of-life, poisoned surface (reactive), or RF match moving power into reflection.

NUMBERS WORTH KNOWING

– Ar sputter: few mTorr · target voltage 300–600 V · yields ~0.5–2 atoms/ion · racetrack utilization ~30% · adhesion layers: Ti, Cr, TiW 5–20 nm.

REFERENCE SHEET

ALD SHEET — ONE LAYER AT A TIME

THE CYCLE

– Dose A → purge → dose B → purge. Both half-reactions saturate → growth-per-cycle fixed (~0.5–1.5 Å/cy) → thickness = GPC × cycles.

– Proofs of ALD-ness: dose-saturation curves flat · purge-saturation flat · T-window plateau · thickness linear through zero · high-AR conformality.

TEMPERATURE WINDOW FAILURES

– Too cold: condensation (GPC↑, impure) or incomplete reaction (GPC↓, impure). Too hot: decomposition (CVD leak-in, GPC↑) or desorption (GPC↓).

PURGE DISCIPLINE

– Under-purge = parasitic CVD: GPC drifts up, particles, uniformity loss. Purge time scales with chamber volume, wall temperature, and structure depth.

– High-AR rule: exposure (dose × time) must saturate the bottom; plasma-ALD radicals recombine downhole — thermal ALD wins in the deepest structures.

COMMON CHEMISTRIES

– Al_2O_3: TMA + H_2O (the model system, ~1.1 Å/cy) · HfO_2: $HfCl_4$ or TDMAH + H_2O/O_3 · TiN: $TiCl_4$ + NH_3 · W: WF_6 + Si_2H_6/B_2H_6 nucleation.

– O_3 or plasma co-reactants: lower T, denser films — at the price of AR reach and possible substrate oxidation.

INTERVIEW ONE-LINERS

– 'ALD trades time for control: flux becomes irrelevant once you saturate.' · 'Count cycles, not seconds.' · 'If GPC depends on dose, it isn't ALD yet.'

REFERENCE SHEET

CVD SHEET — CONTINUOUS GROWTH

TWO REGIMES DECIDE EVERYTHING

– Surface-reaction-limited (lower T): rate = k·e^(–Ea/RT) → temperature uniformity is everything; flow forgiven → batch furnaces.

– Mass-transport-limited (higher T): rate set by boundary-layer delivery → flow/geometry is everything; temperature forgiven → single-wafer reactors.

– Uniformity problem? First question: which regime — check rate sensitivity to T (Arrhenius slope) vs to flow.

VARIANTS AND WHY THEY EXIST

– LPCVD: low P → high diffusivity → conformal, batch (poly-Si, nitride) · APCVD: fast, transport-limited (old oxides).

– PECVD: plasma cracks precursors → 200–400 °C deposition, stress tunable via dual-frequency; H-rich films · HDP-CVD: simultaneous dep+sputter → gap fill.

– MOCVD: metal-organics for compound semis (GaN, InP) — epitaxial CVD with chemistry-limited zoo of parasitics.

FILM-QUALITY LEVERS

– Higher T generally: denser, purer, better conformality (surface mobility) · Plasma power: stress and density at fixed T · Dilution and P: gas-phase nucleation control (particles).

FAILURE SIGNATURES

– Depletion along flow → thickness ramp across load/chamber → raise flow or drop T toward reaction-limited · Particles after clean → seasoning missing · Haze → gas-phase nucleation → lower P or precursor partial pressure.

REFERENCE SHEET

ETCH SHEET — SEVEN KNOBS, FOUR SHAPES

THE SYNERGY SENTENCE

– Radicals weaken chemically, ions drive vertically; together they beat the sum (Coburn–Winters). Sidewalls: chemistry without bombardment → passivation decides the profile.

KNOB TABLE

– Source power → flux/rate · Bias → ion energy: anisotropy, selectivity ↓, damage ↑ · Pressure → ion angle spread + radical:ion ratio.

– F:C ratio (fluorocarbons): O_2 add → leaner, etch-y · H or CH_4 add → richer, polymer-y, selective · Wafer T ↓ → passivation sticks → straighter, slower.

PROFILE → CAUSE → CORRECTION

– Taper: over-passivation / weak ions → lean chemistry, +bias, –pressure · Undercut: passivation failing → +passivant, –T, –pressure.

– Bow: ion scattering/charging mid-depth → –pressure, check mask facet · Notch/foot at stop: charging on insulator → pulsed plasma, lower overetch bias.

– Micro-trench: ion reflection off sidewall · Striations: mask LER or passivation cycling · Grass: micromask (residue, sputtered chamber material).

ENDPOINT & CONTROL

– OES: product lines fall / reactant lines rise at clear · Interferometry: fringe counting = live rate · Always define overetch % and know its stop-layer cost.

– ARDE: narrow etches slower (transport + charging) → bias ↑, P ↓, pulse, or ALE for the critical landing.

REFERENCE SHEET

SPC SHEET — PROVE IT'S UNDER CONTROL

THE THREE QUESTIONS

– Is the gauge real? (GRR <10% of tolerance good; >30% unusable) → never chase drift smaller than gauge noise.

– Is the process stable? (control chart: its own ±3σ voice) · Is it good enough? (capability vs specs) — in that order.

CAPABILITY CHEAT LINES

– Cp = (USL–LSL)/6σ — spread only · Cpk = min(USL–μ, μ–LSL)/3σ — charges for off-center.

– Cpk 1.00 ≈ edge at 3σ (~0.13% one-tail) · 1.33 = 4σ margin (house floor) · 1.67 = 5σ (critical layers). Pp/Ppk: same math, long-term σ.

– Cpk ■ Cp → recenter (cheap win) · Cp itself low → variance hunt (DOE).

CHART SIGNALS (ACT ON PATTERNS, NOT JUST EXCURSIONS)

– 1 pt outside 3σ · 8 in a row one side (shift) · 6 trending (drift: consumable/target life) · 2 of 3 beyond 2σ · cycling (chamber/shift stratification).

– Control limits ≠ spec limits. Never draw specs on the chart to 'control' to them.

DOE CRASH RULES

– Screen with 2-level fractionals + 3–5 center points, randomize, replicate centers · OFAT misses interactions and finds false optima.

– Resolution IV+ so mains aren't aliased with 2-factor interactions · curvature at centers → go response-surface before optimizing.

– Nested variance first: within-wafer vs wafer-to-wafer vs lot-to-lot names the mechanism class before you spend wafers.

KEEP GOING

This pack is the floor, not the ceiling.

One deep-dive lesson lands in your inbox each week — the mechanisms behind these answers, real fab case studies, and the statistics to prove your process is under control. Questions, corrections or war stories: **info@thinfilmengineer.com**.

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